1. Presentation Layer (Storefront Browser)
Web Browser (Customer / Admin / Guest)
HTTP / UI Interaction
⚡ Next.js 14 (React 18 + TypeScript + Tailwind CSS) Port 3002
REST API Calls
Token & Session Persistence
Storefront Modules & Interactive UIs
Store Catalog &amp; 280ms Debounced Search (Live Category Filter &amp; Stock Status Badges)
Click Card / Details Button
Product Detail / Quick View Modal (High-Res Image, Specs, Qty Stepper, Buy Now)
Add Qty / Buy Now
Sliding Cart Drawer (Guest LocalStorage &amp; Authenticated Sync)
Proceed to Checkout
Atomic Checkout Modal (tok_success vs tok_decline Card Simulation)
Orders &amp; Price-Frozen Receipts (Strict Cross-Customer Privacy Isolation)
Admin Fulfillment &amp; Live Stock Replenishment (FSM Terminal State: DELIVERED Lock)
Fast Demo User Persona Switcher (Talha Ahmed / Alice Buyer / Sara / Admin / Guest)
LocalStorage JWT Access/Refresh &amp; Guest Cart
Communication Protocols & Security Boundary
HTTPS / REST API Requests <code>Authorization: Bearer &lt;JWT&gt;</code> Content-Type: application/json
CORS Inspected
CORS Middleware Allowed Origins: http://localhost:3002
Dispatched to DRF
JSON Response (HTTP 200, 201, 400, 403, 404)
2. Application & API Layer (Host Bare-Metal)
Django 5.2 + Django REST Framework Port 8002
Security & Permission Guards
SimpleJWT Authentication (HMAC-SHA256 Token Verification)
Role-Based Access Control (Customer vs Admin Permissions)
Domain ViewSets & Core Services
ProductViewSet <code>/api/products/, /update_stock/</code> <i>Catalog, Search, &amp; Admin Stock Adjust</i>
CartViewSet <code>/api/cart/, /add/, /update_quantity/</code> <i>Multi-Quantity Validation &amp; Auto Subtotal</i>
⚡ Concurrency-Safe Checkout Service <code>transaction.atomic()</code> select_for_update() in Ascending ID Order
MockPaymentGateway <i>Deterministic Tokens &amp; Idempotency</i>
OrderViewSet <code>/api/orders/, /update_status/</code> <i>FSM: pending-&gt;paid-&gt;shipped-&gt;delivered (Terminal)</i>
psycopg2-binary / Row Locks (SELECT FOR UPDATE)
ACID Committed Transactions
3. Persistent Data Layer (PostgreSQL 18)
PostgreSQL 18 Engine Port 5432 | Database: ecommerce_db
Relational Schema (3NF)
users_customuser PK: id | role: customer, admin
products_product PK: id | UQ: sku CHECK (stock &gt;= 0)
cart_cart &amp; cart_cartitem PK: id | FK: user_id, product_id
payments_paymenttransaction PK: id | UQ: idempotency_key
orders_order PK: id | UQ: order_number status: pending, paid, shipped, delivered...
orders_orderitem PK: id | FK: order_id, product_id Frozen unit_price Snapshot
Performance & Integrity Indexes
idx_product_stock (stock)
idx_order_user_created (user_id, created_at DESC)
idx_payment_idempotency (idempotency_key UNIQUE)
idx_cart_user (user_id UNIQUE)